MTCIPv6E - MikroTik Certified IPv6 Engineer

Differences between IPv4 and IPv6. IPv6 address

IPv4
2^32=4294967296

Solutions:
VLSM
NAT/PAT

IPv5: Yes, but it was used for an experimental project

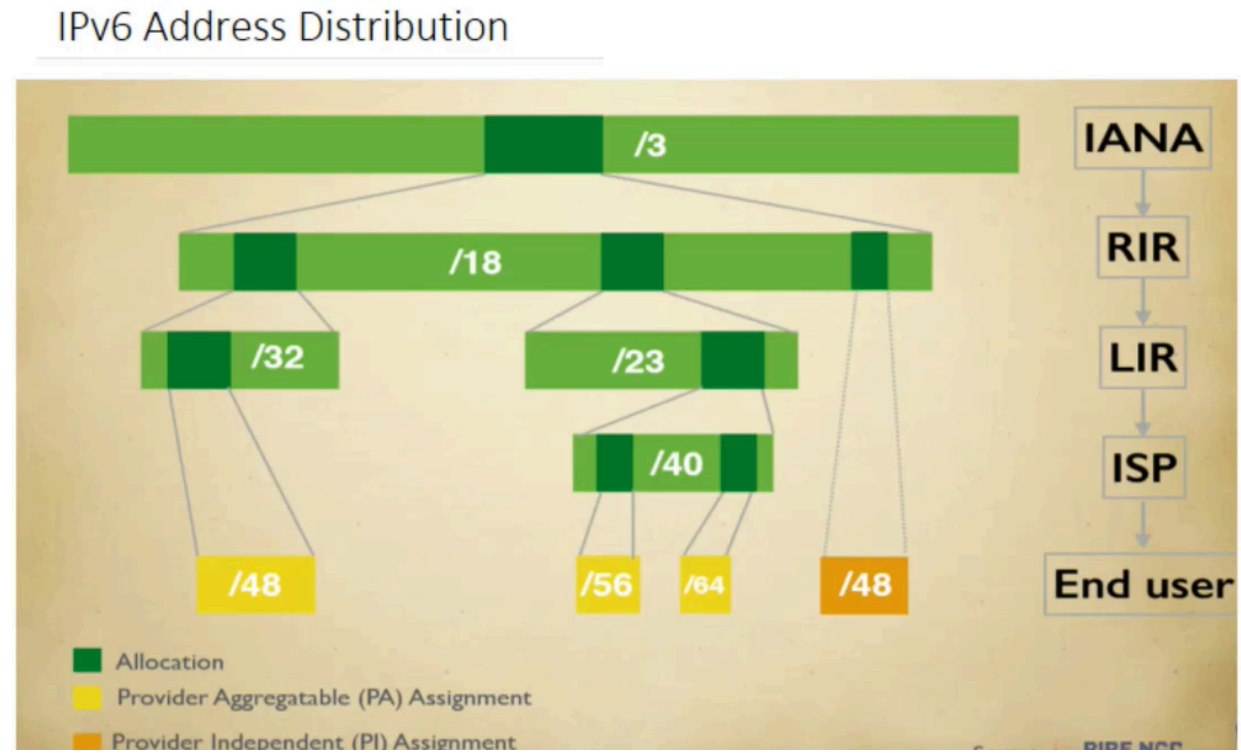
IPv6
128bit
Total-340,282,366,920,938,463,463,374,607,431,768,211,456

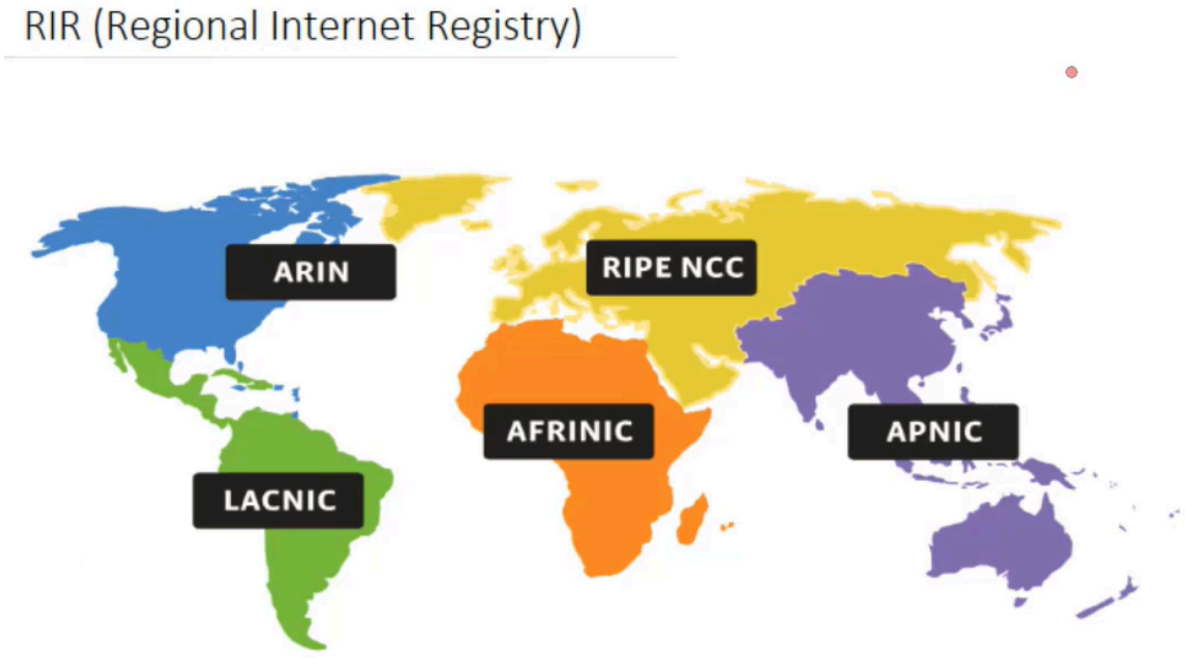
340- undecillion
282- decillion
366- nonillion
920- octillion
938- septillion
463- sextillion
463- quintillion
374- quadrillion
607- trillion
431- billion
768- million
211- thousand
456

IPv6 Characteristics

- No broadcast traffic
- Mobility
- No NAT/PAT
- IPsec always run
- Improved header: Doesn't require checksums
- Migration toll: IPv4 and IPv6 are not compatible so we need migration tools. Such as "6to4" tunnel and "dual stack"

Address distribution





Address notation

IPv6 notation

2001:0db8:0be0:75a2:0000:0000:0000:0001

Leading zeros can be left out

2001:db8:be0:75a2:0:0:0:1

Consecutive fields of zeros can be replaced with ::

2001:db8:be0:75a2::1

IPv6 notation (cont.)

2001:0db8:0000:0000:0010:0000:0000:0001

If there are several consecutive fields of zeros only one can be replaced with ::

2001:db8::10:0:0:1

You can choose which one

2001:db8:0:0:10::1

The same IP address. Both notations are valid but the first one is recommended

IPv6 notation (LAB1)

Compress the following IPv6 addresses to shortest form possible:

2001:0db8:0ab0:0d00:0000:0000:0000:0c01

2001:0db8:0000:4c05:0000:0000:05ad:0bb1

2001:0db8:0000:0000:1234:0000:0000:da61

Prefix length

IPv6 prefix

2001:1234:5678:1234:5678:ABCD:EF12:1234/64



IPv6 prefix (cont.)

Find the prefix of the below IPv6 address:

2001::1234:5678:ABCD:EF12:1234/64

IPv6 prefix (cont.)

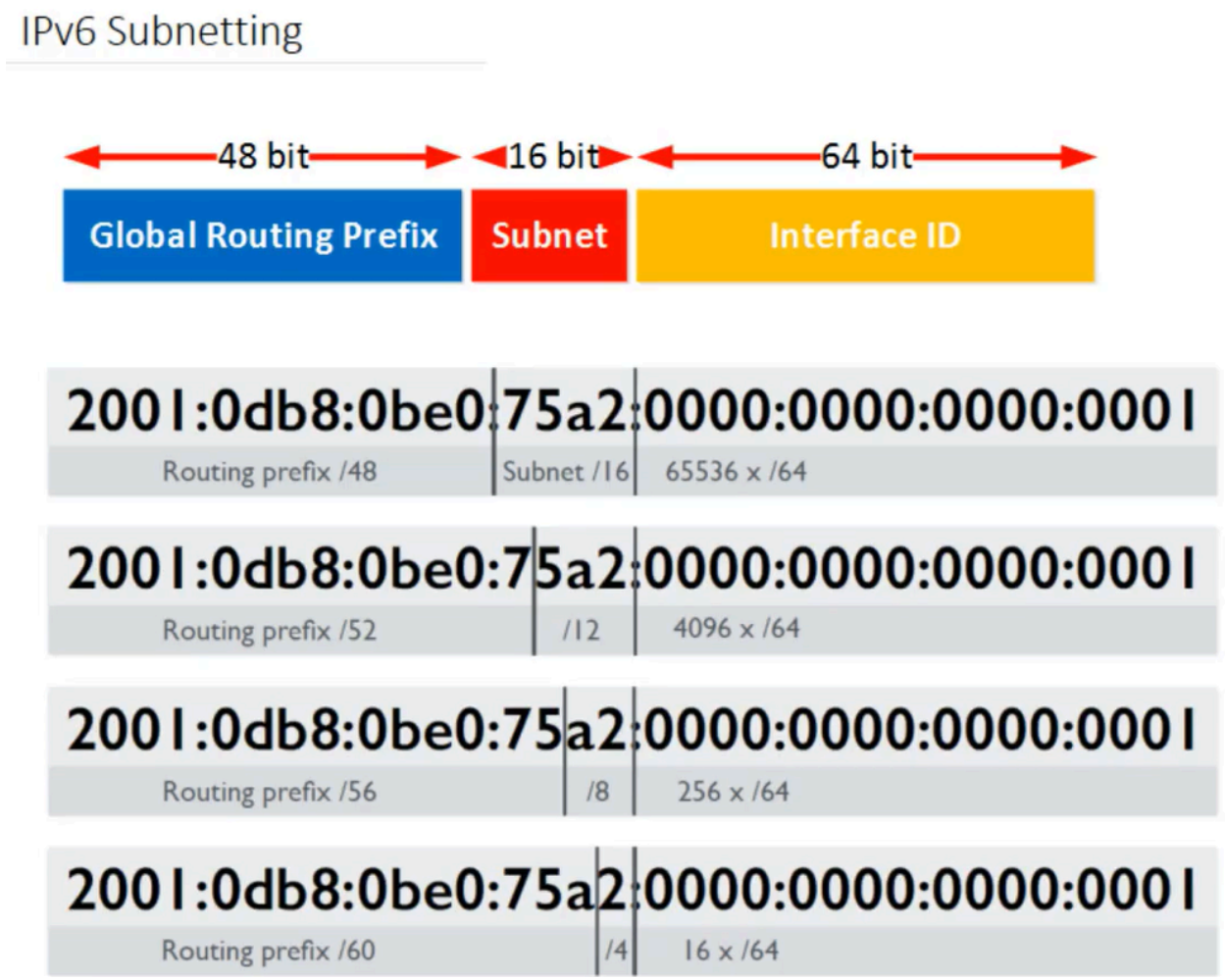
Find the prefix of the below IPv6 address:

2001:1234:abcd:5678:9877:3322:5541:aabb/53

IPv6 address types

IPv6 address types

Type	Range
Link local	fe80::/10
Global unicast	2000::/3
Multicast	ff00::/8
Unique local	fc00::/7



IPv6 package enabling and ping from pc

SLAAC (Mini DHCP server)

If you don't have IPv6 uplink just configure this

Pool-2001:db8:2162:8450::/60

LAN- 2001:db8:2162:8450::1/64

DNS-2001:4860:4860::8888

Check from your PC ensure Ipv6 is enabled in your PC

EUI-64

Traditional interface identifiers for network adapters are 48-bit MAC address. This address consists of a 24-bit manufacturer ID and a 24-bit board ID.

IEEE EUI-64 is a new standard for network interface addressing. The company ID is still 24-bits in length, but the extension ID is 40 bits, creating a much larger address space for network adapters.

To create an EUI-64 address from the interface MAC address:

- 0xFFFE is inserted into the MAC address between the manufacturer ID and the board ID.
- seventh bit of the first byte is reversed.

Lets make an example with following MAC address `00:0C:42:28:79:45`.

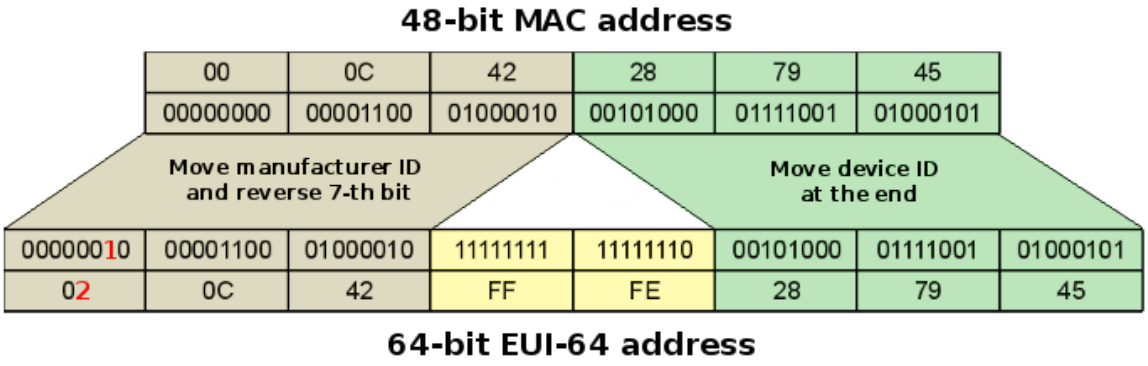


Image above illustrates conversion process. When the result is converted to colon-hexadecimal notation, we get the interface identifier `20C:42FF:FE28:7945`

Subnetting

128bit			
64bit			64bit
Network ID			Host ID
ISP/32bit	Customer/48bit	Subnet/64bit	
32bit	16bit	16bit	

2000:ACAD:1234:6600:0000:0000:0000:0000/56

2000:ACAD:1234:6600::/56

6600

0110 0110 0000 0000

0110 0110/56 0000 0000

2^8=256

Frist 4 subnet

66 0000 0000 = 6600/64

66 0000 0001 = 6601/64

66 0000 0010 = 6602/64

66 0000 0011 = 6603/64

Last 2 subnet

66 1111 1110 = 66FE/64

66 1111 1111 = 66FF/64

IPv6 address types

Check IPv6 address lecture

IPv6 Address Types

Prefix	Designation and Explanation	IPv4 Equivalent
::/128	**Unspecified** This address may only be used as a source address by an initialising host before it has learned its own address.	0.0.0.0
::1/128	**Loopback** This address is used when a host talks to itself over IPv6. This often happens when one program sends data to another.	127.0.0.1
::ffff/96 Example: ::ffff:192.0.2.47	**IPv4-Mapped** These addresses are used to embed IPv4 addresses in an IPv6 address. One use for this is in a dual stack transition scenario where IPv4 addresses can be mapped into an IPv6 address. See RFC 4038 for more details.	There is no equivalent. However, the mapped IPv4 address can be looked up in the relevant RIR's Whois database.
fc00::/7 Example: fdf8:f53b:82e4::53	**Unique Local Addresses (ULAs)** These addresses are reserved for local use in home and enterprise environments and are not public address space. These addresses might not be unique, and there is no formal address registration. Packets with these addresses in the source or destination fields are not intended to be routed on the public Internet but are intended to be routed within the enterprise or organisation. See RFC 4193 for more details.	Private, or RFC 1918 address space: 10.0.0.0/8 172.16.0.0/12 192.168.0.0/16
fe80::/10 Example: fe80::200:5aee:feaa:20a2	**Link-Local Addresses** These addresses are used on a single link or a non-routed common access network, such as an Ethernet LAN. They do not need to be unique outside of that link. Link-local addresses may appear as the source or destination of an IPv6 packet. Routers must not forward IPv6 packets if the source or destination contains a link-local address. Link-local addresses may appear as the source or destination of an IPv6 packet. Routers must not forward IPv6 packets if the source or destination contains a link-local address.	169.254.0.0/16

This sheet is available at www.ripe.net/ipv6-address-types • Produced by the RIPE NCC in cooperation with ICANN • www.ripe.net • www.icann.org

IPv6 Address Types

Prefix	Designation and Explanation	IPv4 Equivalent
2001:0000::/32 Example: 2001:0000:4136:e378: 8000:63bf:3fff:fdd2	**Teredo** This is a mapped address allowing IPv6 tunneling through IPv4 NATs. The address is formed using the Teredo prefix, the server's unique IPv4 address, flags describing the type of NAT, the obfuscated client port and the client IPv4 address, which is probably a private address. It is possible to reverse the process and identify the IPv4 address of the relay server, which can then be looked up in the relevant RIR's Whois database. You can do this on the following webpage: **http://www.potaroo.net/cgi-bin/ipv6addr**	No equivalent
2001:0002::/48 Example: 2001:0002:6c::430	**Benchmarking** These addresses are reserved for use in documentation. They should not be used as source or destination addresses.	198.18.0.0/15
2001:0010::/28 Example: 2001:10:240:ab::a	**Orchid** These addresses are used for a fixed-term experiment. They should only be visible on an end-to-end basis and routers should not see packets using them as source or destination addresses.	No equivalent
2002::/16 Example: 2002:cb0a:3cdd:1::1	**6to4** A 6to4 gateway adds its IPv4 address to this 2002::/16, creating a unique /48 prefix. As the IPv4 address of the gateway router is used to compose the IPv6 prefix, it is possible to reverse the process and identify the IPv4 address, which can then be looked up in the relevant RIR's Whois database. You can do this on the following webpage: **http://www.potaroo.net/cgi-bin/ipv6addr**	There is no equivalent but 192.88.99.0/24 has been reserved as the 6to4 relay anycast address prefix by the IETF.
2001:db8::/32 Example: 2001:db8:8:4::2	**Documentation** These addresses are used in examples and documentation. They should never be source or destination addresses.	192.0.2.0/24 198.51.100.0/24 203.0.113.0/24
2000::/3	**Global Unicast** Other than the exceptions documented in this table, the operators of networks using these addresses can be found using the Whois servers of the RIRs listed in the registry at: **http://www.iana.org/assignments/ipv6-unicast-address-assignments**	No equivalent single block
ff00::/8 Example: ff01:0:0:0:0:0:0:2	**Multicast** These addresses are used to identify multicast groups. They should only be used as destination addresses, never as source addresses.	224.0.0.0/4

This sheet is available at www.ripe.net/ipv6-address-types • Produced by the RIPE NCC in cooperation with ICANN • www.ripe.net • www.icann.org

Address configuration

Auto-configuration

IPv6 Stateless Autoconfiguration

All interfaces on IPv6 nodes must have a link-local address, which is usually automatically configured from the identifier for an interface and the link-local prefix FE80::/10. A link-local address enables a node to communicate with other nodes on the link and can be used to further configure the node.

Stateless – SLAAC, DHCPv6

Stateless Address Autoconfiguration(SLAAC)

SLAAC is a method in which the host or router interface is assigned a 64-bit prefix, and then the last 64 bits of its address are derived by the host or router with help of EUI-64 process. SLAAC uses NDP protocol to work.

Stateless DHCPv6 to provide other information, such as DNS server, domain name, SIP server, to the clients. Stateless DHCPv6 Service for IPv6 is sometimes called DHCPv6 Lite.

ipv6 protocol Stateful – DHCPv6

The two types of autoconfiguration are "stateless" and "stateful." Stateful autoconfiguration is the IPv6 equivalent of DHCP. A new protocol, called DHCPv6 (and based closely on DHCP), is used to pass out addressing and service information in the same way that DHCP is used in IPv4.

This protocol is a stateful counterpart to "IPv6 Stateless Address Autoconfiguration" (RFC 2462), and can be used separately,

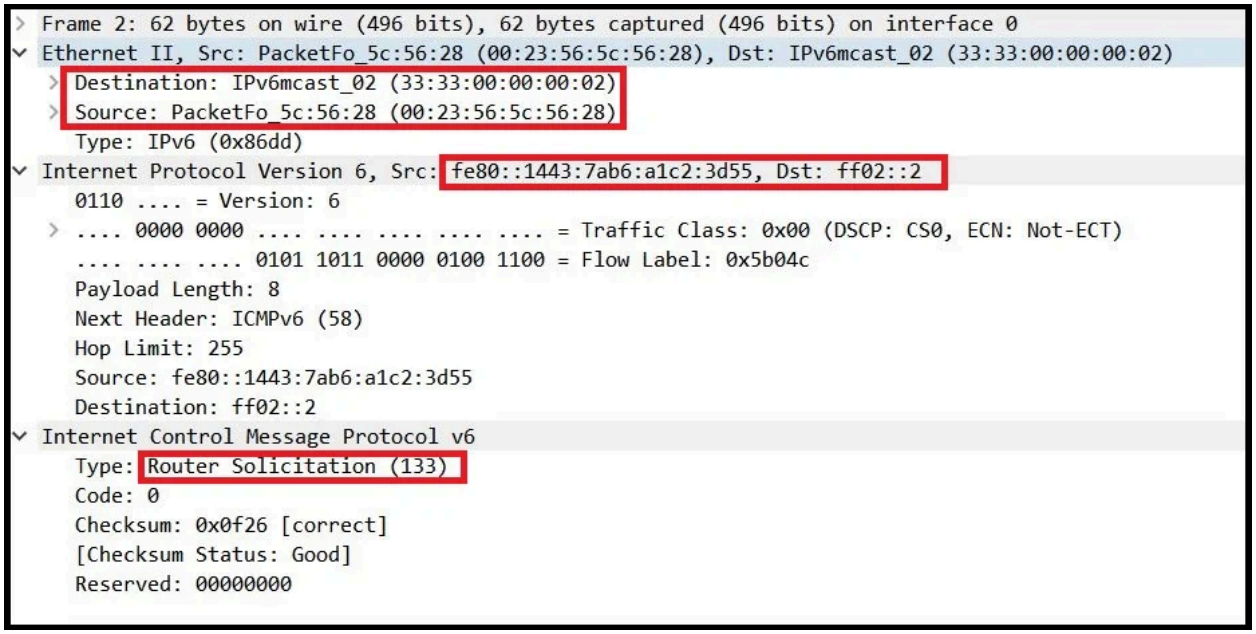
Neighbor discovery protocol

The Neighbor Discovery Protocol (NDP) is a protocol used with Internet Protocol Version 6 (IPv6). It operates at the link layer of the Internet model and is responsible for gathering various information required for internet communication, including the configuration of local connections and the domain name servers and gateways used to communicate with more distant systems.

NDP defines five ICMPv6 packet types for the purpose of router solicitation, router advertisement, neighbor solicitation, neighbor advertisement, and network redirects.

Router Solicitation (Type 133)

Hosts inquire with Router Solicitation messages to locate routers on an attached link.[3] Routers which forward packets not addressed to them generate Router Advertisements immediately upon receipt of this message rather than at their next scheduled time.



Router Advertisement (Type 134)

Routers advertise their presence together with various link and Internet parameters either periodically, or in response to a Router Solicitation message.

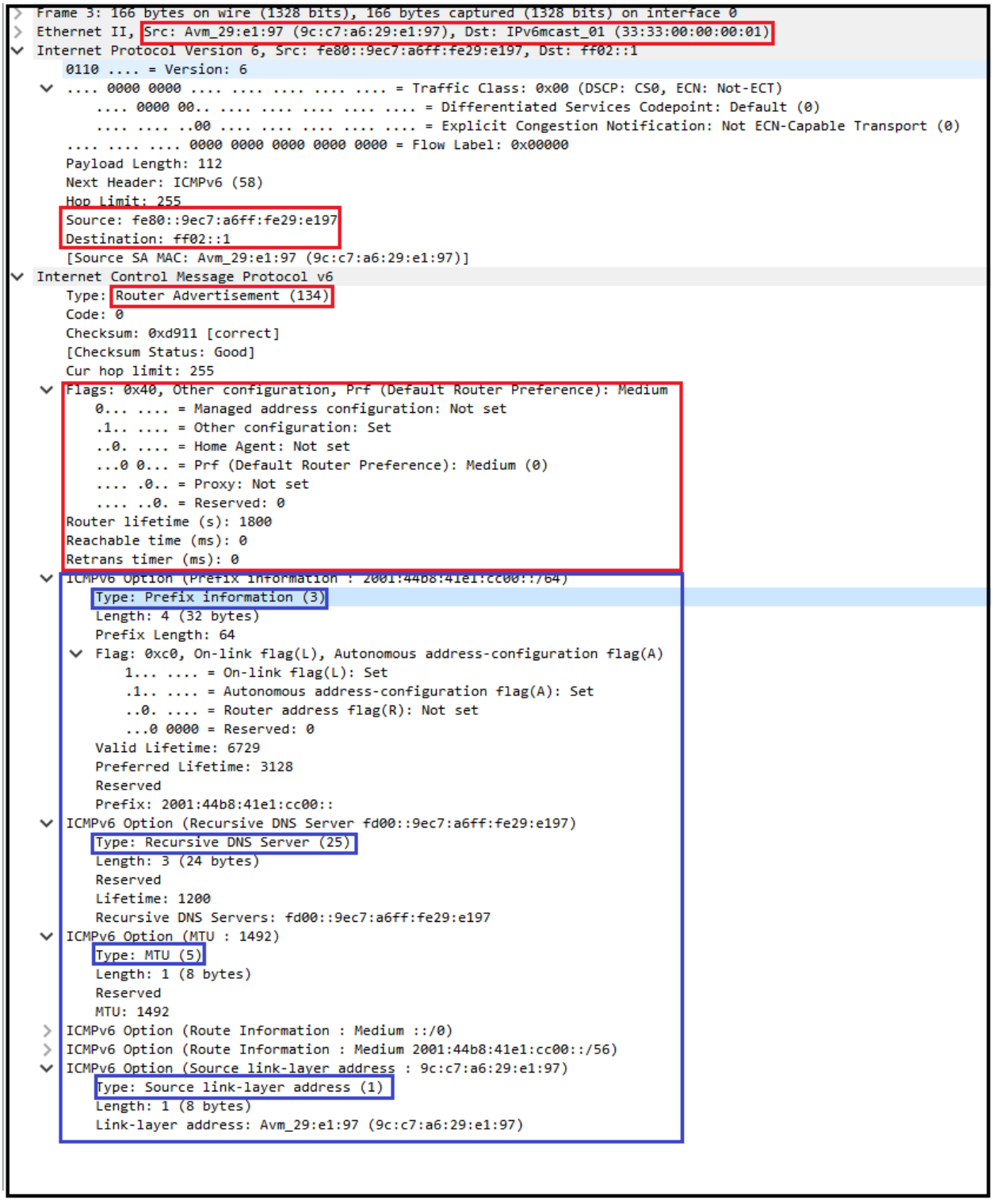
Here is the meaning of certain flags:

Managed (M flag) = A ‘1’ means the address is provided by stateful DHCPv6.

Other (O flag) = A ‘1’ means that the address is provided by Stateless DHCPv6, useful for providing options when the client is doing SLAAC–Stateless Address Autoconfiguration. (Note, there is no option for a default gateway).

Address Autoconfiguration (A flag) = A ‘1’ means that it suggests the host creates its own IPv6 address using SLAAC.

Router lifetime = anything other than a ‘0’ value means the source sending the RA will be the default gateway. (Note, it is a link-local address and not global unique address).



Neighbor Solicitation (Type 135)

Neighbor solicitations are used by nodes to determine the link layer address of a neighbor, or to verify that a neighbor is still reachable via a cached link layer address. Also
Duplicate Address Detection

325002: Duplicate address ipv6_address/MAC_address on interface

Neighbor Advertisement (Type 136)

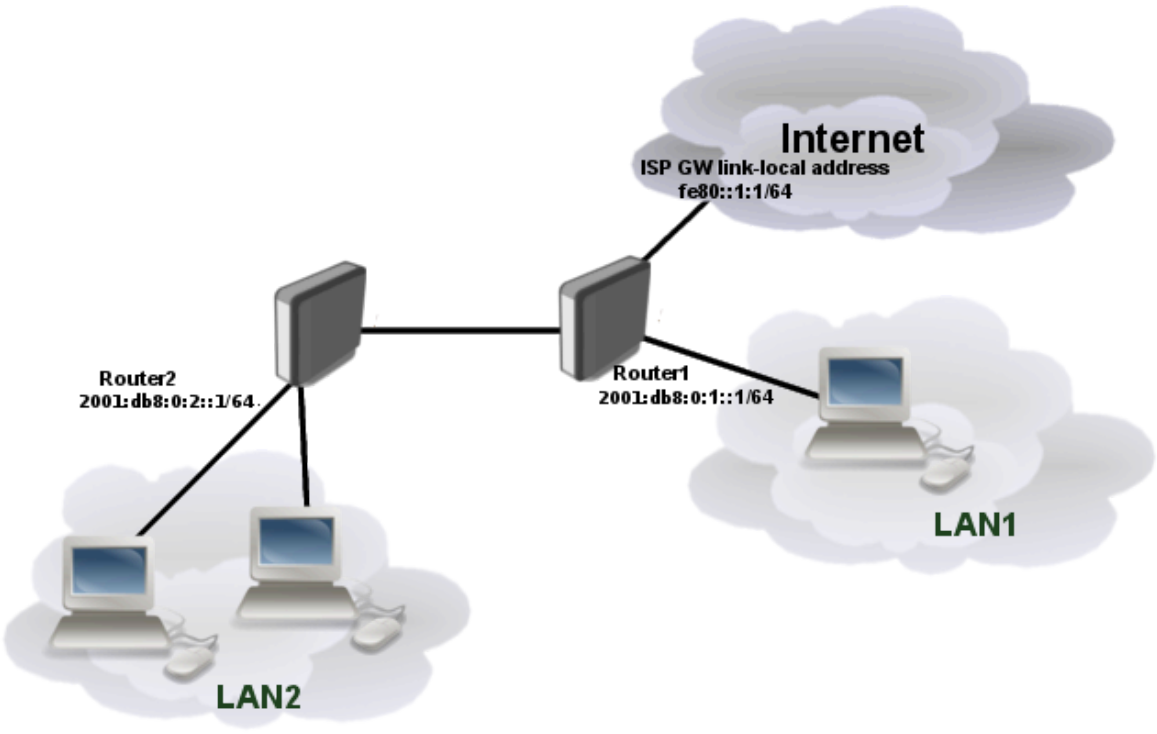
Neighbor advertisements are used by nodes to respond to a Neighbor Solicitation message.

Redirect (Type 137)

Routers may inform hosts of a better first hop router for a destination

IPv6 routing basics

Static routing



Router1

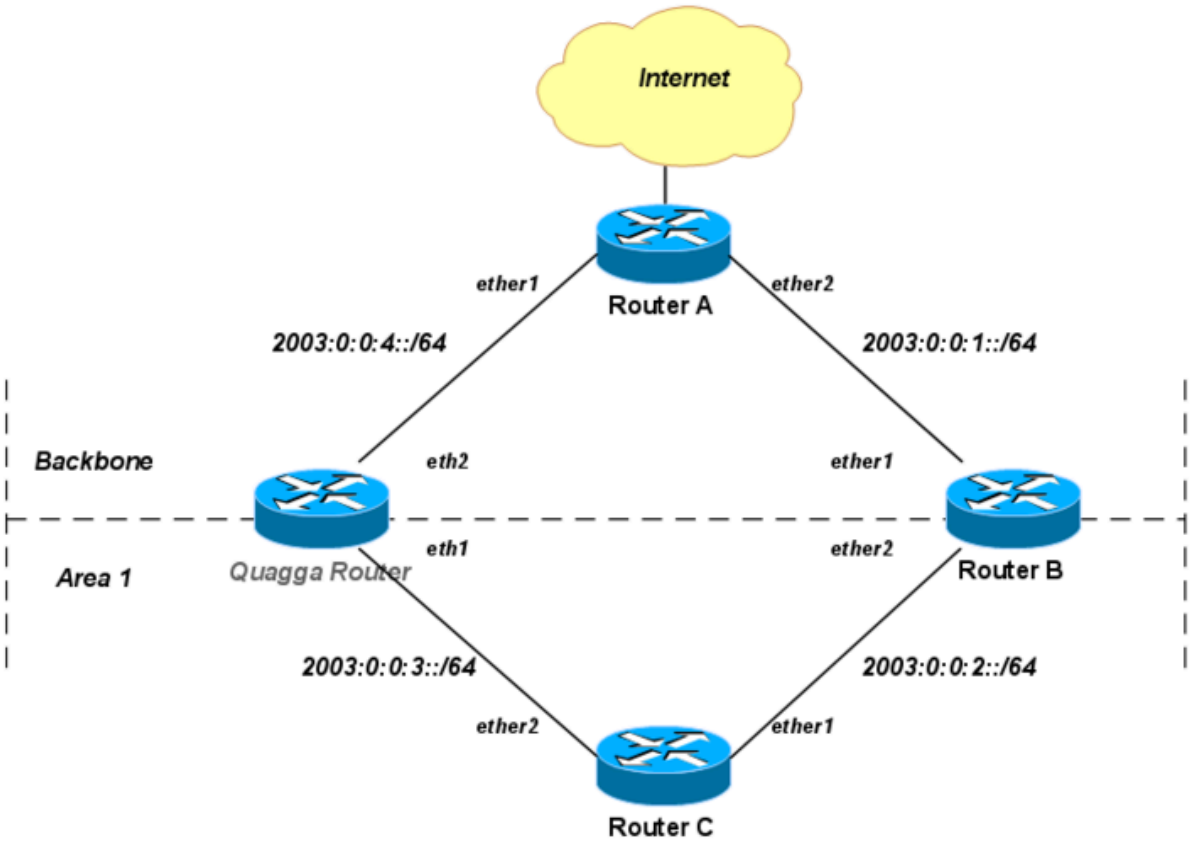
/ipv6 address
add address=2001:db8:0:1::1/64 interface=ether3 advertise=yes

/ipv6 route
add gateway=fe80::1:1%ether1
add dst-address=2001:db8:0:2::/64 gateway=fe80::219:d1ff:fe39:3535%ether2

Router2
/ipv6 address
add address=2001:db8:0:2::1/64 interface=ether2 advertise=yes

/ipv6 route
add gateway=fe80::219:d1ff:fe00:3512%ether1

OSPF



Router-1
/ipv6 address
 add address=2003::1:0:0:0:1/64 advertise=no interface=ether2
 add address=2003::4:0:0:0:1/64 advertise=no interface=ether1
 add address=2003::1/64 advertise=no interface=ToInternet

/routing ospf-v3
 set router-id=0.0.0.1 distribute-default=always-as-type-1

/routing ospf-v3 interface
 add interface=ether1 area=backbone
 add interface=ether2 area=backbone

Router-2
/ipv6 address
 add address=2003::1:0:0:0:2/64 advertise=no interface=ether1
 add address=2003::2:0:0:0:2/64 advertise=no interface=ether2

/routing ospf-v3
 set router-id=0.0.0.2

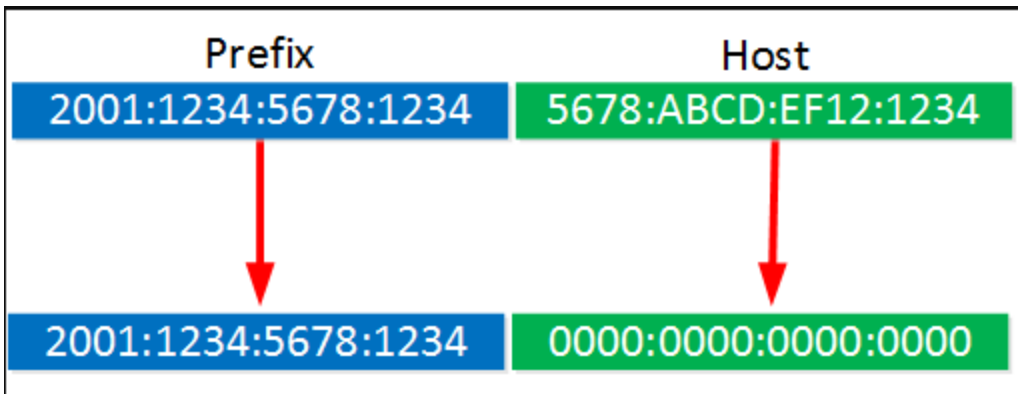
```
/routing ospf-v3 area
 add area-id=0.0.0.1 name=area1
/routing ospf-v3 interface
 add interface=ether1 area=backbone
 add interface=ether2 area=area1
```

Configure special router

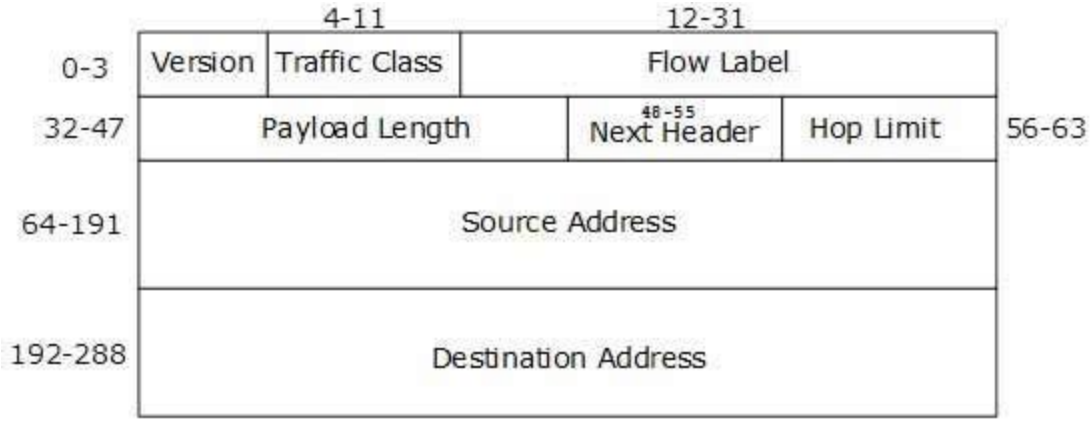
```
Router-3
/ipv6 address
 add address=2003::2:0:0:0:3/64 advertise=no interface=ether1
 add address=2003::3:0:0:0:3/64 advertise=no interface=ether2

/routing ospf-v3
 set router-id=0.0.0.3
/routing ospf-v3 area
 add area-id=0.0.0.1 name=area1
/routing ospf-v3 interface
 add interface=ether1 area=area1
 add interface=ether2 area=area1
```

ipv6 prefix



IPv6 header



IPv6 fixed header is 40 bytes long and contains the following information.

S.N. Field & Description

1

Version (4-bits): It represents the version of Internet Protocol, i.e. 0110.

2

Traffic Class (8-bits): These 8 bits are divided into two parts. The most significant 6 bits are used for Type of Service to let the Router Known what services should be provided to this packet. The least significant 2 bits are used for Explicit Congestion Notification (ECN).

3

Flow Label (20-bits): This label is used to maintain the sequential flow of the packets belonging to a communication. The source labels the sequence to help the router identify that a particular packet belongs to a specific flow of information. This field helps avoid re-ordering of data packets. It is designed for streaming/real-time media.

4

Payload Length (16-bits): This field is used to tell the routers how much information a particular packet contains in its payload. Payload is composed of Extension Headers and Upper Layer data. With 16 bits, up to 65535 bytes can be indicated; but if the Extension Headers contain Hop-by-Hop Extension Header, then the payload may exceed 65535 bytes and this field is set to 0.

5

Next Header (8-bits): This field is used to indicate either the type of Extension Header, or if the Extension Header is not present then it indicates the Upper Layer PDU. The values for the type of Upper Layer PDU are same as IPv4's.

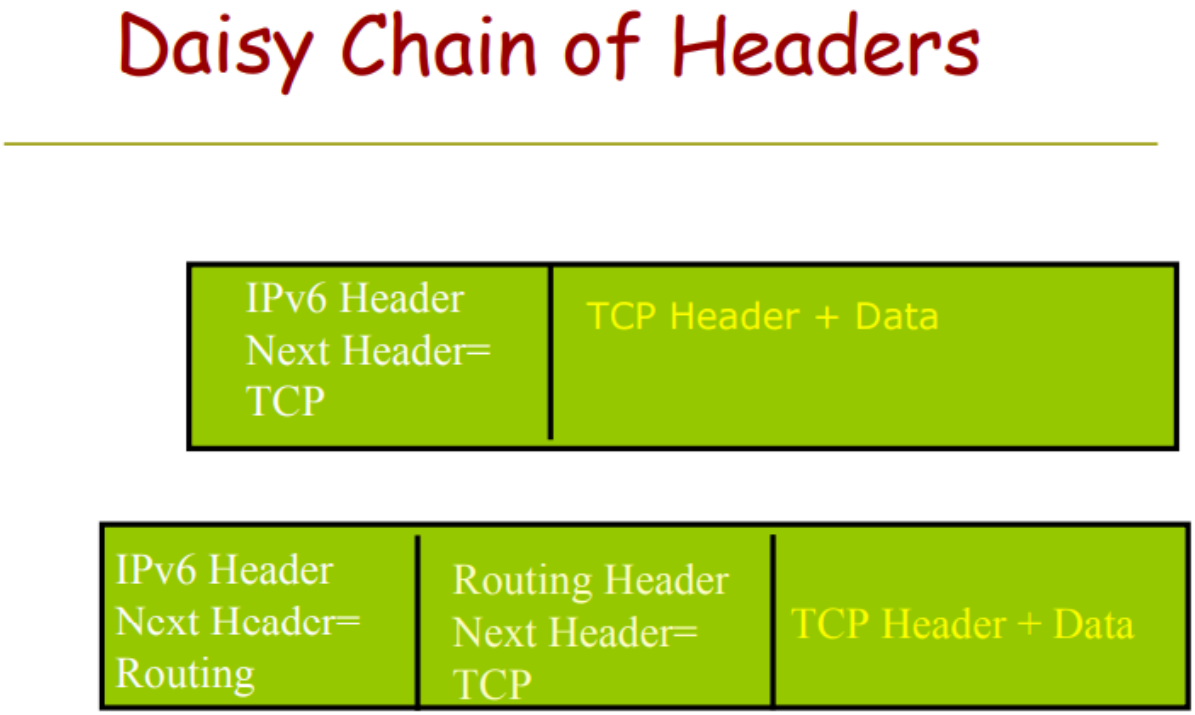
6

Hop Limit (8-bits): This field is used to stop packet to loop in the network infinitely. This is same as TTL in IPv4. The value of Hop Limit field is decremented by 1 as it passes a link (router/hop). When the field reaches 0 the packet is discarded.

7
Source Address (128-bits): This field indicates the address of originator of the packet.

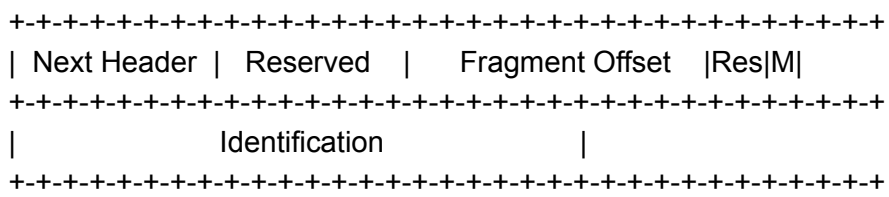
8
Destination Address (128-bits): This field provides the address of intended recipient of the packet.

Next header (daisy chaining)



Fragmentation

The Fragment header is used by an IPv6 source to send a packet larger than would fit in the path MTU to its destination. (Note: unlike IPv4, fragmentation in IPv6 is performed only by source nodes, not by routers along a packet's delivery path -- see section 5.) The Fragment header is identified by a Next Header value of 44 in the immediately preceding header, and has the following format:



Next Header 8-bit selector. Identifies the initial header type of the Fragmentable Part of the original packet (defined below). Uses the same values as the IPv4 Protocol field [RFC-1700 et seq.].

Reserved 8-bit reserved field. Initialized to zero for transmission; ignored on reception.

Fragment Offset 13-bit unsigned integer. The offset, in 8-octet units, of the data following this header, relative to the start of the Fragmentable Part of the original packet.

Res 2-bit reserved field. Initialized to zero for transmission; ignored on reception.

M flag 1 = more fragments; 0 = last fragment.

Identification 32 bits.

Path MTU discovery
Path MTU Discovery (PMTUD) is a standardized technique in computer networking for determining the maximum transmission unit (MTU) size on the network path between two Internet Protocol (IP) hosts, usually with the goal of avoiding IP fragmentation.

Initially, Internet Control Messaging Protocol (ICMP) was introduced to aid network troubleshooting by providing tools used to verify end-to-end reachability. ICMP also reports back errors on hosts. Unfortunately, due to its nature and lack of built-in security, it quickly became target for many attacks. For example, ICMP REQUESTS are used by an attacker for network reconnaissance. ICMP lack of inherent security opened it up to a number of vulnerabilities. Resulting in security teams blocking all ICMP message types. Having adverse effects on useful ICMP features such as Path MTU.

ICMPv6(IPv6 Security)
ICMP for v4 and v6 are completely different. Unlike ICMP for IPv4, ICMPv6 is an integral part of IPv6 communication and ICMPv6 has features that are required for IPv6 operation. For this reason its not possible to simply block ICMPv6 and all its message types. ICMPv6 is a legitimate part of V6; need to select what you can filter.

Prevent ICMPv6 Address Spoofing
Best practice to check source and destination address in an ICMPv6 packet. For example, MLD (Multicast Listener Discovery), the source should always be a link-local address. If this is not the case, it is likely the packet has originated from an illegal source and should be dropped. You may also block any ICMPV6 address that has not been assigned by the IANA. However, this is a manual process and ACL adjustment are made whenever IANA makes changes to the list.

MLD (Multicast Listener Discovery)
Multicast Listener Discovery (MLD) is a component of the Internet Protocol Version 6 (IPv6) suite. MLD is used by IPv6 routers for discovering multicast listeners on a directly attached link, much like Internet Group Management Protocol (IGMP) is used in IPv4. The protocol is embedded in ICMPv6 instead of using a separate protocol. MLDv1 is similar to IGMPv2 and MLDv2 similar to IGMPv3.

Temporary addresses
The IPv6 Privacy Extension is defined in RFC 4941. It is a format defining temporary addresses that change in regular time intervals; successive addresses appear unrelated to each other for outsiders and are a means of protection against address correlation

Firewall

Disable IPv6 Neighbour Discovery

/ipv6 nd set [find] disabled=yes

IPv6 firewall to a router

work with new packets, accept established/related packets;
drop link-local addresses from Internet interface;
accept access to a router from link-local addresses, accept multicast addresses for management purposes, accept your address for router access;
drop anything else;

```
/ipv6 firewall filter
add action=accept chain=input comment="allow established and related"
connection-state=established,related
add chain=input action=accept protocol=icmpv6 comment="accept ICMPv6"
add chain=input action=accept protocol=udp port=33434-33534 comment="defconf: accept
UDP traceroute"
add chain=input action=accept protocol=udp dst-port=546 src-address=fe80::/16
comment="accept DHCPv6-Client prefix delegation."
add action=drop chain=input in-interface=sit1 log=yes log-prefix=dropLL_from_public
src-address=fe80::/16
add action=accept chain=input comment="allow allowed addresses" src-address-list=allowed
add action=drop chain=input
/ipv6 firewall address-list
add address=fe80::/16 list=allowed
add address=xxxx::/48  list=allowed
add address=ff02::/16 comment=multicast list=allowed
```

IPv6 firewall for clients

Enabled IPv6 puts your clients available for public networks, set proper firewall to protect your customers.

accept established/related and work with new packets;
drop invalid packets and put prefix for rules;
accept ICMP packets;
accept new connection from your clients to the Internet;
drop everything else.

```
/ipv6 firewall filter
add action=accept chain=forward comment=established,related
connection-state=established,related
add action=drop chain=forward comment=invalid connection-state=invalid log=yes
log-prefix=ipv6,invalid
add action=accept chain=forward comment=icmpv6 in-interface=!sit1 protocol=icmpv6
```

add action=accept chain=forward comment="local network" in-interface=!sit1 src-address-list=allowed
add action=drop chain=forward log-prefix=IPV6

IPsec
IPsec is a group of protocols that are used together to set up encrypted connections between devices. It helps keep data sent over public networks secure. IPsec is often used to set up VPNs, and it works by encrypting IP packets, along with authenticating the source where the packets come from.

Within the term "IPsec," "IP" stands for "Internet Protocol" and "sec" for "secure." The Internet Protocol is the main routing protocol used on the Internet; it designates where data will go using IP addresses. IPsec is secure because it adds encryption* and authentication to this process.

Header only encryption (AH)/ Authentication Header
Authentication Header (AH) is a member of the IPsec protocol suite. AH ensures connectionless integrity by using a hash function and a secret shared key in the AH algorithm. AH also guarantees the data origin by authenticating IP packets. Optionally a sequence number can protect the IPsec packet's contents against replay attacks,[17] using the sliding window technique and discarding old packets.

***Authentication Header* format**

Offsets	$Octet_{16}$	0								1								2								3							
$Octet_{16}$	Bit_{10}	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
0	0	*Next Header*								*Payload Len*								*Reserved*															
4	32	*Security Parameters Index (SPI)*																															
8	64	*Sequence Number*																															
C	96	*Integrity Check Value (ICV)*																															
...	...	...																															

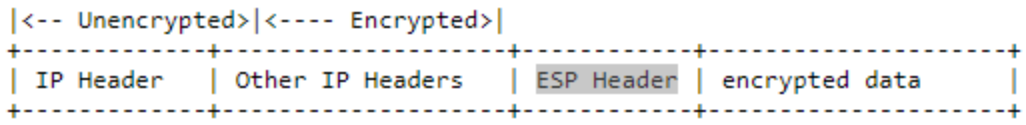
Data only encryption (ESP)
Encapsulating Security Payload (ESP). The ESP protocol provides data confidentiality (encryption) and authentication (data integrity, data origin authentication, and replay protection). ESP can be used with confidentiality only, authentication only, or both confidentiality and authentication.

Header and data encryption (AH+ESP)
IPSec uses two distinct protocols, Authentication Header (AH) and Encapsulating Security Payload (ESP), which are defined by the IETF.

The AH protocol provides a mechanism for authentication only. AH provides data integrity, data origin authentication, and an optional replay protection service. Data integrity is ensured by using a message digest that is generated by an algorithm such as HMAC-MD5 or HMAC-SHA.

Data origin authentication is ensured by using a shared secret key to create the message digest. Replay protection is provided by using a sequence number field with the AH header. AH authenticates IP headers and their payloads, with the exception of certain header fields that can be legitimately changed in transit, such as the Time To Live (TTL) field.

The ESP protocol provides data confidentiality (encryption) and authentication (data integrity, data origin authentication, and replay protection). ESP can be used with confidentiality only, authentication only, or both confidentiality and authentication. When ESP provides authentication functions, it uses the same algorithms as AH, but the coverage is different. AH-style authentication authenticates the entire IP packet, including the outer IP header, while the ESP authentication mechanism authenticates only the IP datagram portion of the IP packet.



Transition Mechanisms

1. Dual stack
2. 6to4
3. 6RD
4. Teredo
5. DS-lite (Dual stack lite)

Dual Stack
Fully functional IPv4 and IPv6 work side by side.
The most recommended way of implementing IPv6
Also endorsed by RIPE

6to4
Allows IPv6 packets to be transmitted over an IPv4 network
IPv6 packets are encapsulated in IPv4 packets
Delivered to a 6to4 relay via IPv4 network
Decapsulated and sent forward as IPv6 packets
Intended only as a transition mechanism, not as a permanent solution

6RD
IPv6 Rapid Deployment is 6to4 derivative
IPv6 relay is controlled by your ISP
From client to ISP is IPv4 network only
On the client side additional software is needed
to encapsulate IPv6 into IPv4 packets

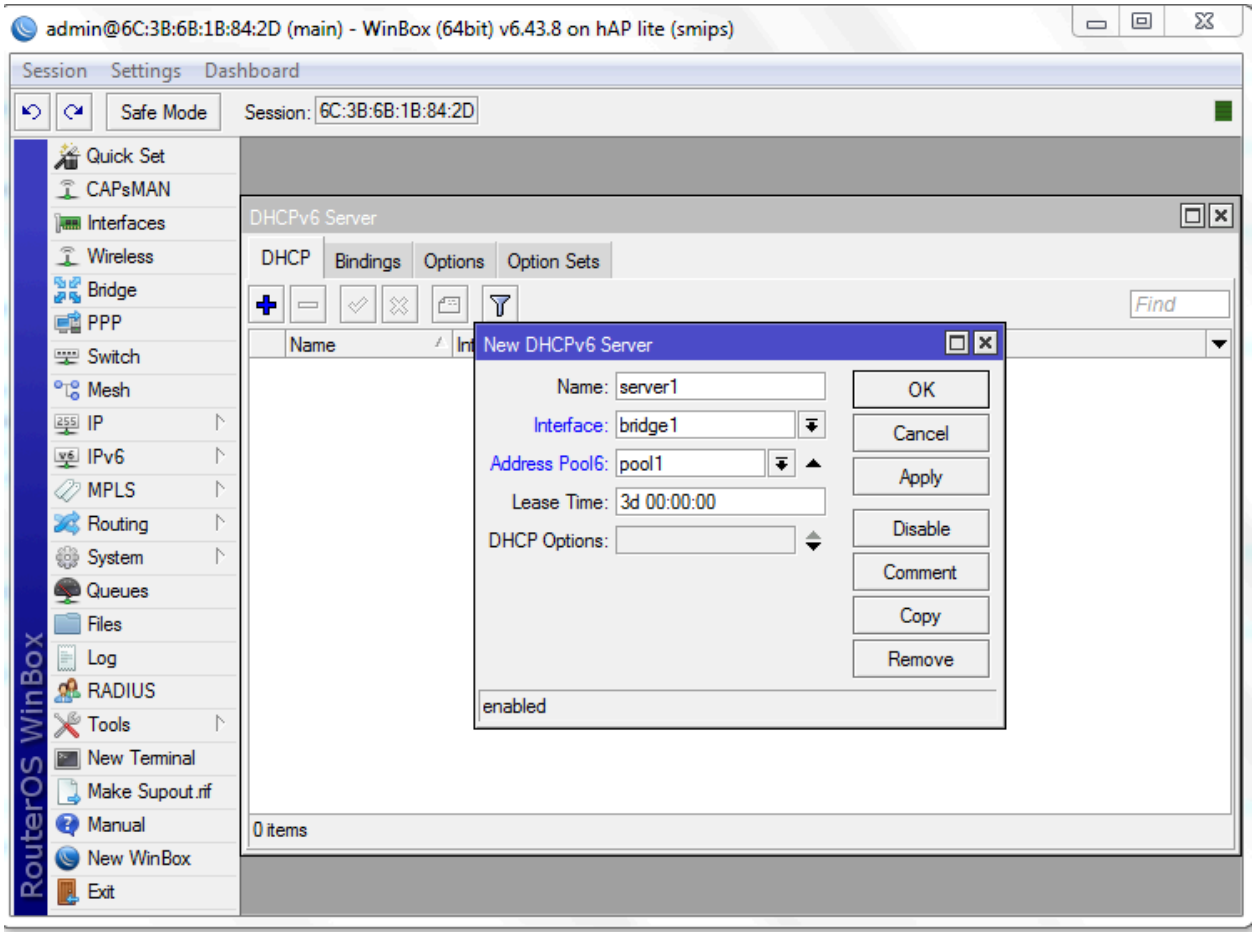
Teredo
Teredo encapsulates IPv6 traffic into IPv4 UDP packets

The traffic is sent through IPv4 Internet
Unlike 6to4, Teredo works behind an IPv4 NAT
Uses Teredo prefix 2001::/32
Can only provide a single IPv6 address per tunnel endpoint
Cannot be used to distribute addresses to
multiple hosts like 6to4
Developed by Microsoft

DS-lite Dual stack lite
IPv6 only links are used between the ISP and the client
Client has native IPv6 connectivity
When and IPv4 packet needs to be sent, it is encapsulated into an IPv6 packet
Sent to the ISP's NAT box which decapsulates and forwards it as IPv4 traffic
NAT is centralized at the ISP level
Clients use private IPv4 addresses (e.g. 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16)
ISP → Client network is IPv6 only

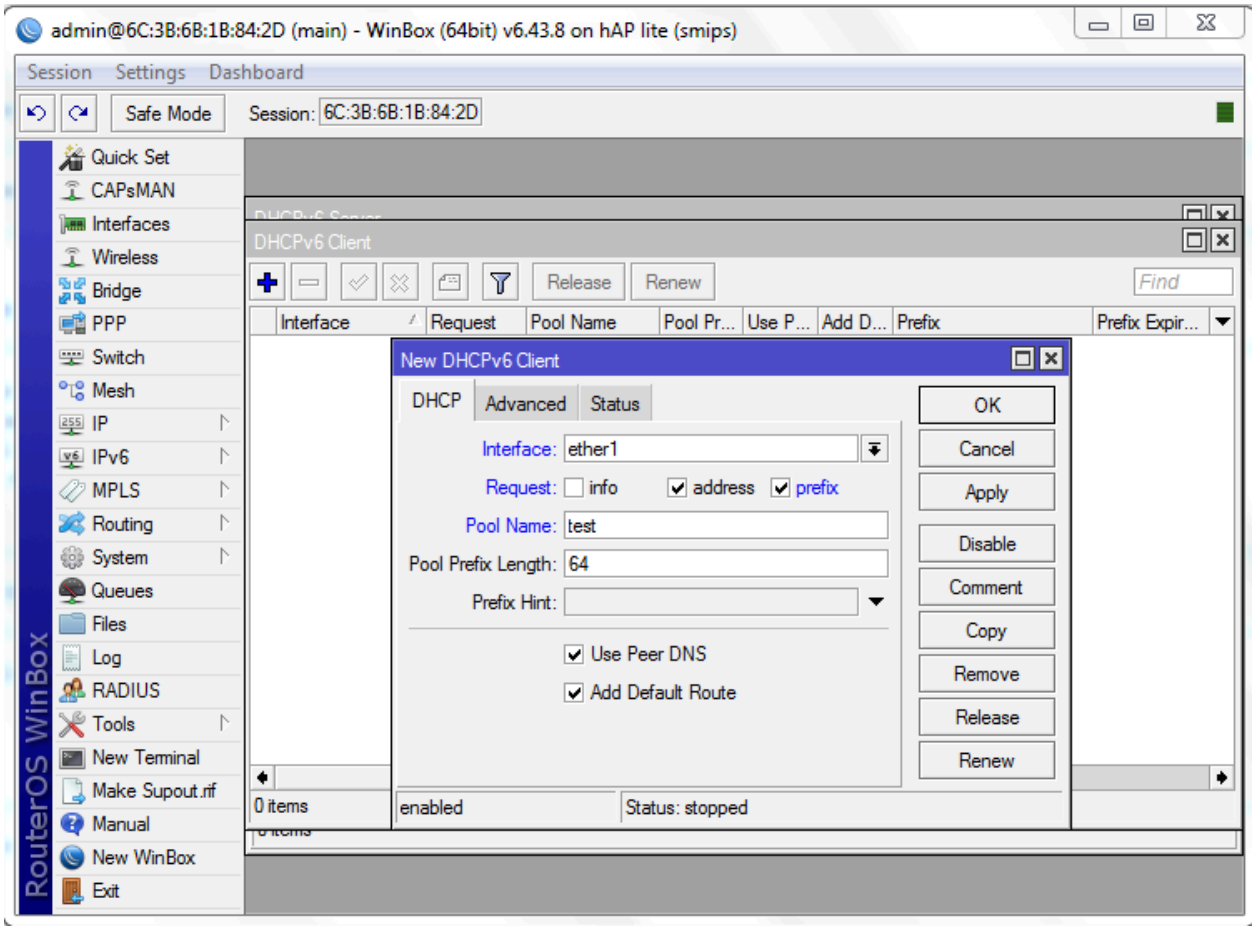
Interoperability
IPv6 pool
ipv6 pool> add
name: test
prefix: 2001::/60
prefix-length: 62

DHCP PD server replace to **DHCP server**
/ipv6 dhcp-server add name=myServer address-pool=myPool interface=local



DHCP-client

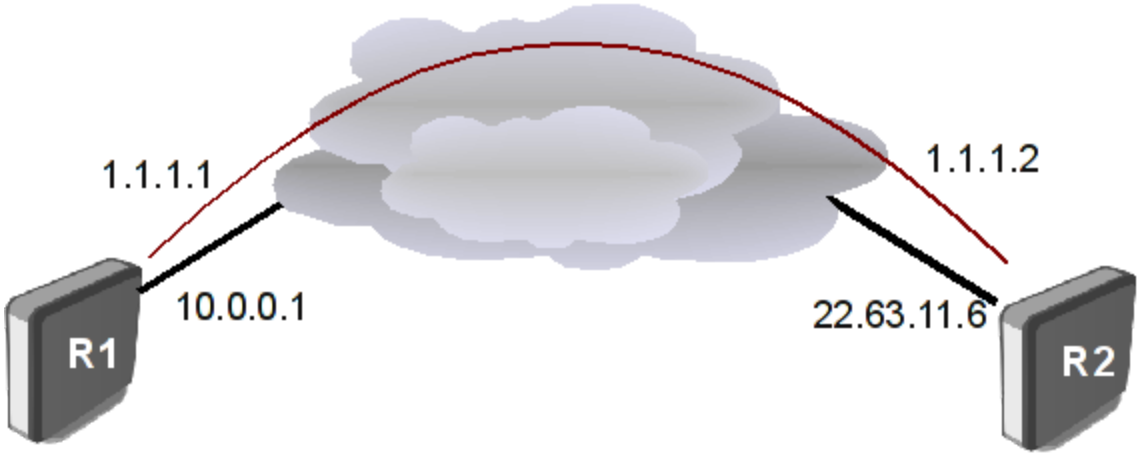
/ipv6 dhcp-client add request=prefix pool-name=test-ipv6 pool-prefix-length=64 interface=ether13



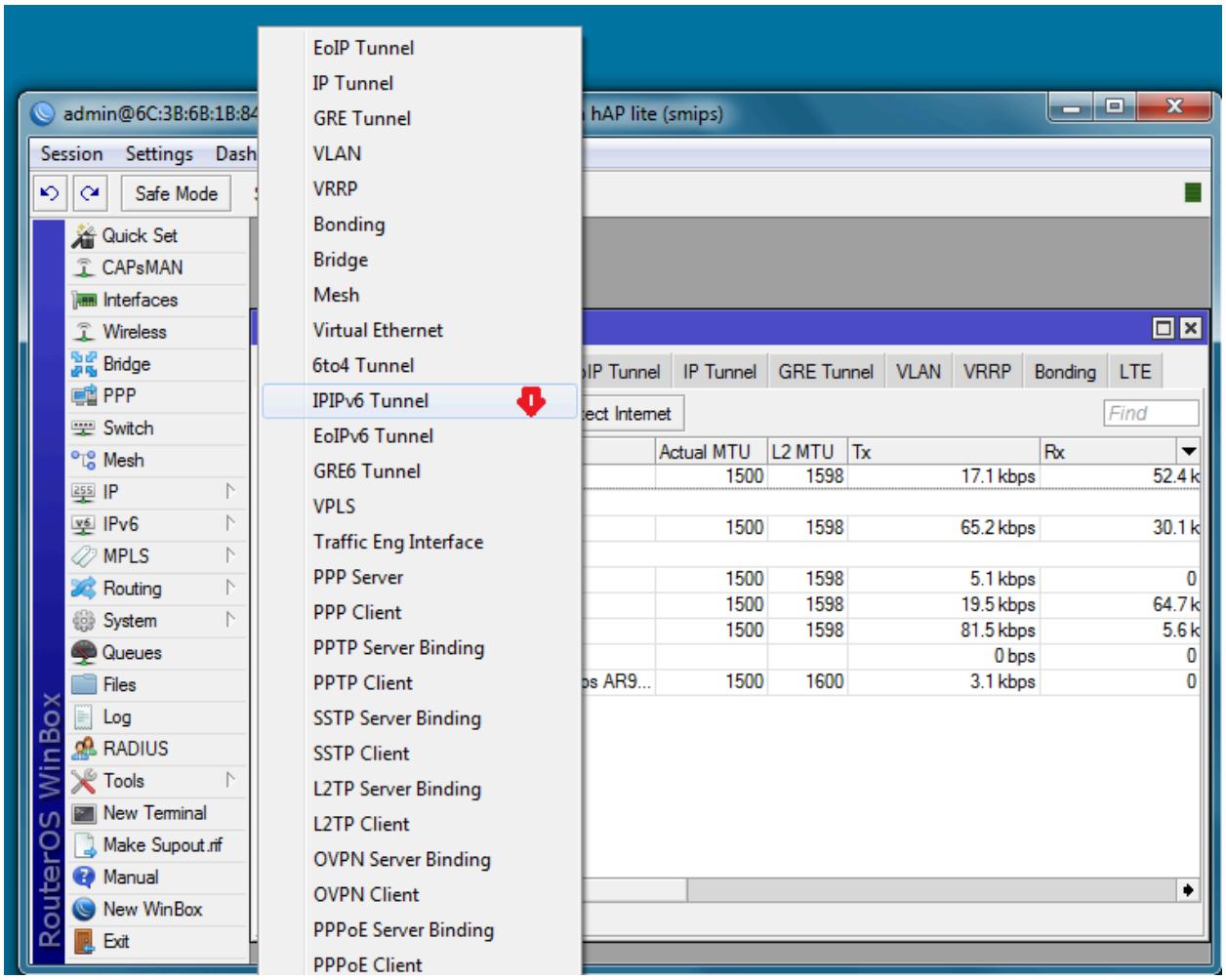
IPv6 tunnels(IPIPv6)

IPIP tunnel is a simple protocol that encapsulates IP packets in IP to make a tunnel between two routers. The IPIP tunnel interface appears as an interface under the interface list. Many routers, including Cisco and Linux, support this protocol. This protocol makes multiple network schemes possible.

Suppose we want to add an IPIP tunnel between routers R1 and R2:

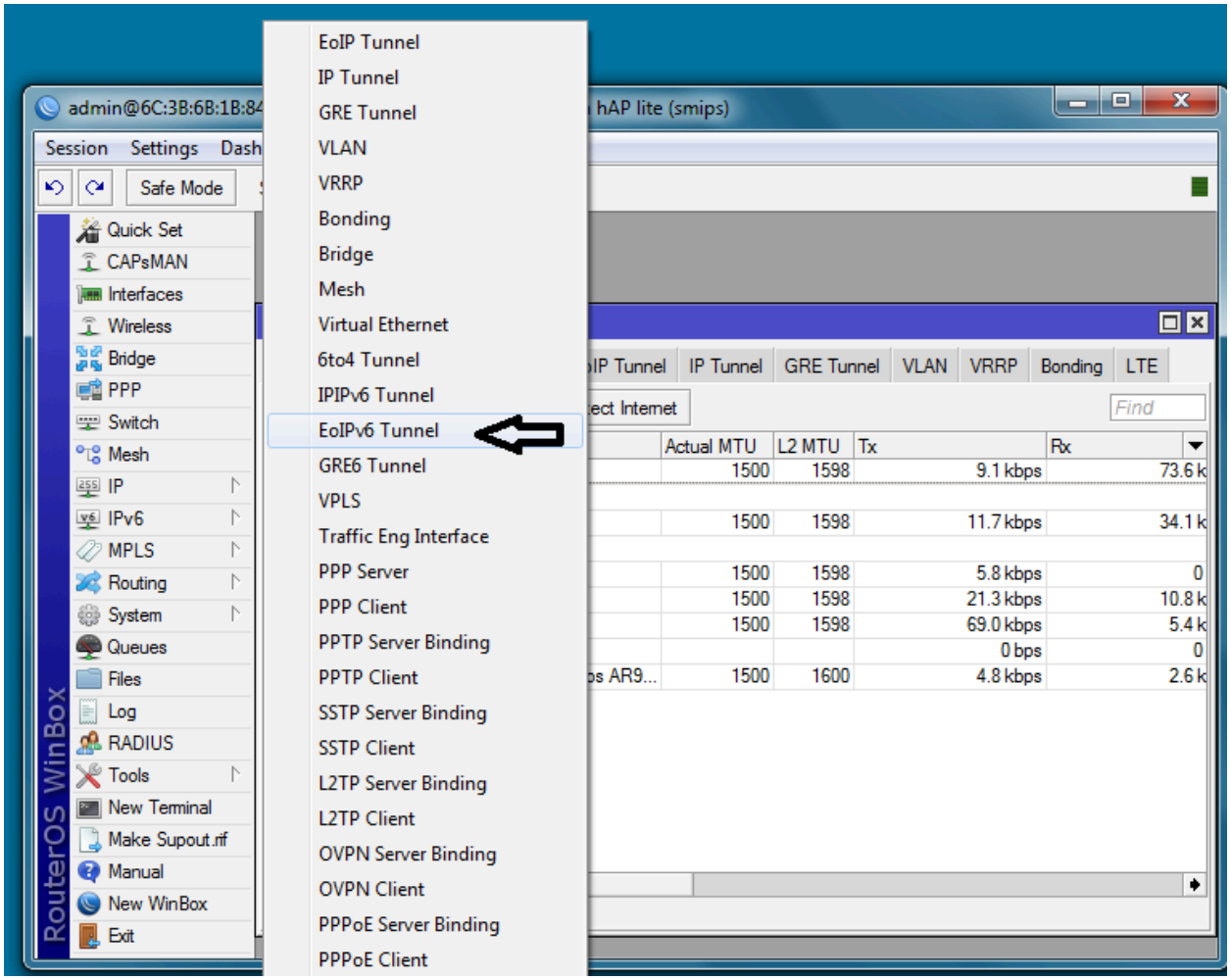


R1
[admin@MikroTik] interface ipip> add
local-address: 10.0.0.1
remote-address: 22.63.11.6



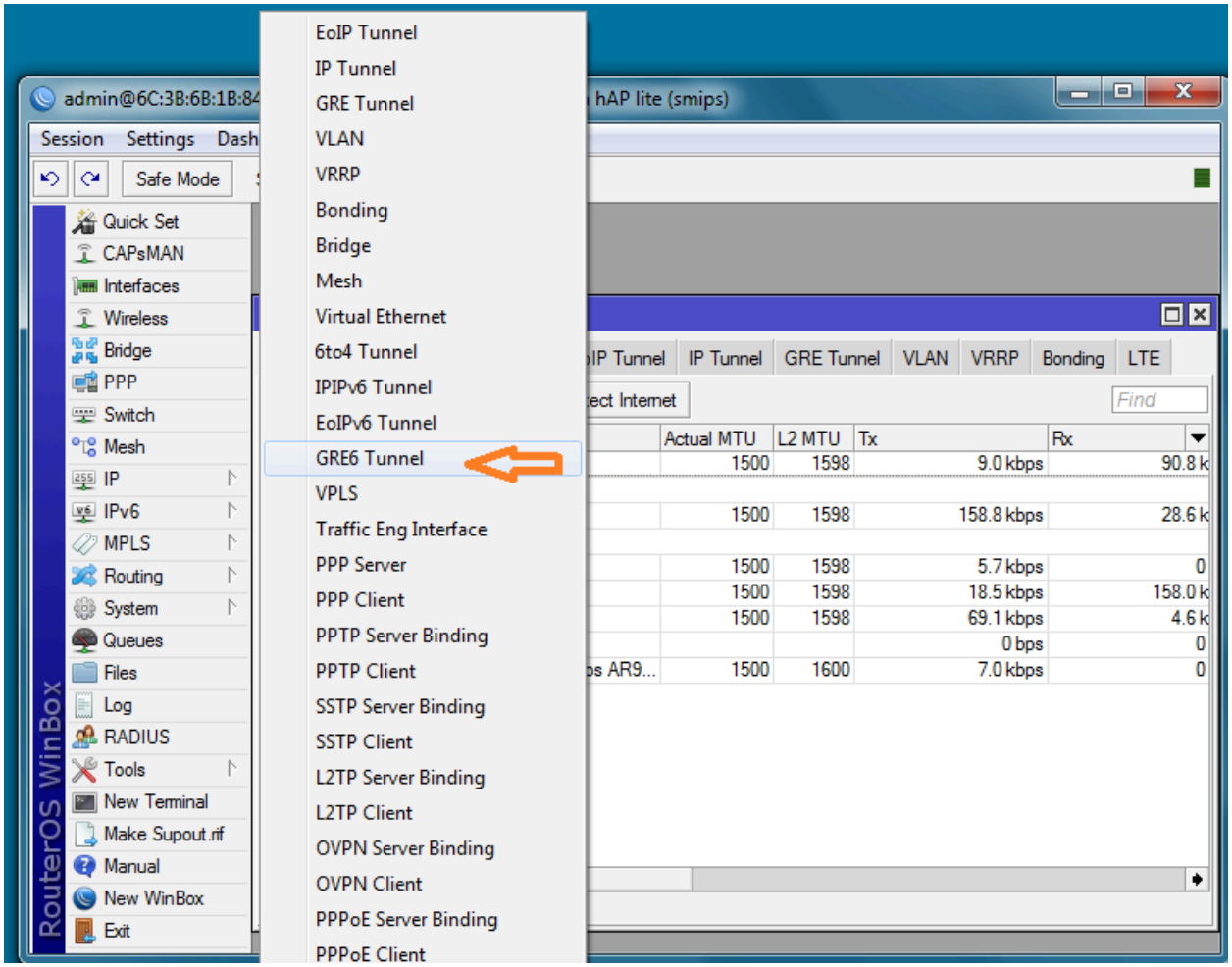
R2
[admin@MikroTik] interface ipip> add
local-address=22.63.11.6
remote-address=10.0.0.1

EoIPv6



GRE6

Interface gre6 is a Generic Routing Encapsulation (GRE) tunnel over IPv6 network. It can encapsulate IPv4 or IPv6 network packets and transfer them over the tunnel.



R1
/interface gre6 add local-address=2001:db8:bad:ee1::a remote-address=2001:db8:f00:baa::b keepalive=5

R2
/interface gre6 add local-address=2001:db8:f00:baa::b remote-address=2001:db8:bad:ee1::a keepalive=5

DNS
Dmain Name System
Forward DNS
Name to IP

Reverse DNS

IP to Name

NTP
Network Time Protocol (NTP).
1.europe.pool.ntp.org
2.europe.pool.ntp.org

PPP IPv6 support
An IPv6 only PPP session is an alternative to the PPP dual stack model; in this scenario a customer gets IPv6 only connectivity using the PPP Session.
This model can be used to allow hosts inside the LAN to establish their own IPv6 PPP sessions, while still receiving IPv4 connectivity from the Residential Gateway (RG). The RG in such a case would not need to be upgraded to support IPv6, although it would need to bridge PPPoE. Another use for this model would be to allow existing BNGs to continue to support IPv4, while new BNGs are used for IPv6 PPP sessions.
https://wiki.mikrotik.com/wiki/Manual:IPv6_PD_over_PPP

global addresses
2000::/3 addresses to the global pool

link-local addresses
A link-local address is an IPv6 unicast address that can be automatically configured on any interface using the link-local prefix FE80::/10

Tools

MikroTik routers provide various tools and utilities for managing and troubleshooting IPv6 networks. Here are some common tools and their usage examples for IPv6:

Ping (IPv6):

Usage: Ping is used to test the reachability of a remote IPv6 host and measure the round-trip time.

Example: To ping an IPv6 address, use the following command:

/ping ipv6-address

Traceroute (IPv6):

Usage: Traceroute is used to trace the route packets take to reach a destination, showing the IPv6 hops along the way.

Example: To trace the route to an IPv6 address, use the following command:

/traceroute ipv6-address

Torch (IPv6):

Usage: Torch is a real-time traffic monitoring tool. It allows you to see IPv6 traffic passing through a specific interface.

Example: To torch IPv6 traffic on a specific interface, use the following command:

/torch interface=interface-name protocol=ipv6

Traffic Generator (IPv6):

Usage: MikroTik routers allow you to generate IPv6 traffic for testing and debugging purposes.

Example: You can create an IPv6 traffic stream for testing purposes in tools like Packet Generator.

Email (IPv6):

Usage: The email tool is used to send email notifications or reports. It can be configured to work with IPv6-enabled email servers.

Example: To configure email settings for sending notifications, use the following command:

/tool email

Netwatch (IPv6):

Usage: Netwatch is used for monitoring hosts or IP addresses and can be used with IPv6 addresses for monitoring.

Example: To create a Netwatch script for IPv6 monitoring, you would use a script like this:

/tool netwatch

add host=ipv6-address interval=10s up-script=up-script-name down-script=down-script-name

Traffic Flow (IPv6):

Usage: Traffic Flow is a feature that allows you to monitor network traffic and collect data about IPv6 traffic flows.

Example: To enable and configure Traffic Flow for IPv6, you would use the /ip traffic-flow settings. The configuration may vary depending on your specific requirements.